

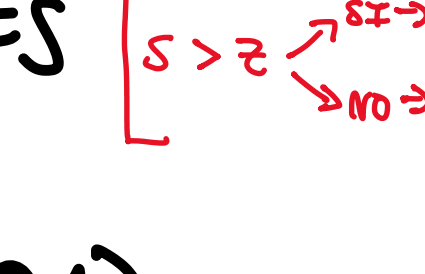
Supervisor:

- * AI was fast and non codifiable knowledge
- * AI can give the firm the same AI knowledge
- * ---

Production Problem:

- * Each human has 1 unit of time and knowledge $z \in [0,1]$
- * A production opportunity has difficulty $x \sim U[0,1]$.

* 1 output if $x \leq z$: $\Pr(x \leq z) = z$



* If $x > z$, it asks a solver (Worker $\approx$ Junior, Solver $\approx$ Senior) with knowledge $s \geq z$:

$$\Pr(\text{Ask for}) = 1-z, \Pr(\text{Success}) = s$$

if $x < z \rightarrow$ evaluate
 $s > z \rightarrow$ resolve
no $\rightarrow$ no resolve

- * One help request consumes $h \in (0,1)$ units of solver time even if the solver fails.

* $n(z)$ identical workers:

$$E[\text{Help Requests}] = n(z)(1-z)$$

Total Expected time of Solver $\left\{ \begin{aligned} &n(z)(1-z) = 1 \\ &\text{Normalization of Solver's time (1 unit)} \end{aligned} \right.$

$$n(z) = \frac{1}{h(1-z)}$$

$\uparrow z \rightarrow \uparrow n(z)$
Cada worker pide menos ayuda y el solver puede atender a más workers

Firms:

$$\pi_1^{\text{Human}}(z) = z - \overset{\text{Expected Output}}{w(z)} \quad \text{Cost}$$

$$\pi_2^{\text{Human}}(s, z) = \underbrace{n(z)s}_{\text{Production Under Solver}} - \underbrace{n(z)w(z)}_{\text{Payment to workers}} - \underbrace{w(s)}_{\text{Payment to solver}}$$

$$\pi_2^{\text{Human}}(s, z) = n(z)[s - w(z)] - w(s)$$

Competitive Equilibria:

- * Firms choose highest π .
- * $\uparrow$ Firms $\Rightarrow (\pi^* \rightarrow 0)$.
- * Workers choose highest w .
- * Compute is allocated at common price r .
- * Human and compute clear their markets.
- * Workers sort:

$$w < I < s$$

* Team output: $n(z)s$

PRE AI WAGE SCHEDULE:

Worker z and matched Solver $m(z)$

$m(z) \equiv$ [The Knowledge of a solver assigned to a worker of knowledge z]
(Before this was labeled 's')

$$\pi_2^{\text{Human}}(s, z) = n(z)s - n(z)w(z) - w(s)$$

Free entry $\Rightarrow (\pi_2^* \rightarrow 0)$

$$\pi_2^{\text{Human}}(m(z), s) = n(z)m(z) - n(z)w(z) - w(m(z)) = 0$$

$$\rightarrow m(z) = w(z) + \frac{w(m(z))}{n(z)}$$

$$w(z) = m(z) - \frac{w(m(z))}{n(z)}$$

Worker wage Schedule before AI

Independent workers make $w(z) = z$.

$$\frac{\partial \pi_2^{\text{Human}}}{\partial m(z)} = n(z) - w'(m(z)) \rightarrow w'(m(z)) = n(z)$$

Solver wage schedule

AI WAGE SCHEDULE:

$$\Pr(\text{AI solver} | x > z_{AI}) = 0 \quad \left[\begin{aligned} &z_{AI} \text{ is the general} \\ &\text{Knowledge of all} \\ &\text{AI's} \end{aligned} \right]$$

Fully Automated Firm:

- * $E[\text{Output}] = z_{AI}$
- * $\pi_1^{\text{AI}} = z_{AI} - r = 0$ (Cost of AI) (No workers)
- * Given Free entry of firms $(\pi_1^{\text{AI}})^* = 0 \rightarrow z_{AI} = r$

Human Workers + AI Solver:

* IA $\rightarrow n(z)$, $z \leq z_{AI}$
Solver Human Workers Workers Knowledge

$$\pi_2^{\text{Top AI}}(z) = \underbrace{n(z)z_{AI}}_{\text{Output of } n(z) \text{ human workers under AI solver}} - \underbrace{n(z)w(z)}_{\text{Cost of } n(z) \text{ human workers}} - \underbrace{r}_{\text{Cost of AI}}$$

$$\pi_2^{\text{Top AI}}(z) = n(z)[z_{AI} - w(z)] - r$$

AI workers + Human Solver:

* 1 human $\rightarrow n(z_{AI})$, $s \geq z_{AI}$
Solver AI workers Solver's Knowledge

$$n(z_{AI}) = \frac{1}{h(1-z_{AI})} \equiv \text{No. of AI workers}$$

$$\pi_2^{\text{Bot AI}}(s) = \underbrace{n(z_{AI})s}_{\text{Output of } n(z_{AI}) \text{ AI workers under human solver}} - \underbrace{n(z_{AI})r}_{\text{Cost of } n(z_{AI}) \text{ AI worker}} - \underbrace{w(s)}_{\text{Cost of Human Solver}}$$

$$\pi_2^{\text{Bot AI}}(s) = n(z_{AI})[s - r] - w(s)$$

THE PROBLEM:

The firm has to choose between:

$$\text{FOOL OF CHOICES} \left\{ \begin{aligned} &\pi_1^{\text{Human}} \\ &\pi_2^{\text{Human}} \\ &\pi_1^{\text{AI}} \\ &\pi_2^{\text{Top AI}} \\ &\pi_2^{\text{Bot AI}} \end{aligned} \right.$$

Note: There are no AI workers + solvers as all AIs share the same Knowledge z_{AI}

Also, each human must end as either:

- * I $\equiv$ Independent Producer
- * W_p = Human worker assisted by people
- * W_a = Human worker assisted by AI
- * S_p = Human solver assisting people
- * S_a = Human solver assisting AI

COMPUTE DISTRIBUTION:

$m \equiv$ Resources available for use

1 compute unit $\rightarrow$ 1 agentic AI

$$m_i + m_w + m_s = m$$

Compute for Independent AI, Compute for AI workers, Compute for AI solvers

We assume there are individuals with knowledge $z \in [0,1]$, such that $z \sim \theta(z)$, CDF(z) = $G(z)$

$g(z)$ is the density function.

We know that:

$$n(z) = \frac{1}{h(1-z)} \rightarrow \text{Use of AI per worker} \rightarrow h(1-z) = \frac{1}{n(z)}$$

Given this, we can define:

$$m_s = \int_{w_a} h(1-z) \theta(z) dz$$

$$m_s = \int_{w_a} h(1-z) dG(z) \quad \frac{dG(z)}{dz} = g(z)$$

Now for m_v , we know that:

$$m_v = n(z_{AI}) \int_{s_a} dG(s)$$

1 solver uses $n(z_{AI})$ workers, Aggregation of all human solvers

Matching Function:

$$m: W_p \rightarrow S_p$$

$m(z)$: Knowledge of a solver assigned to worker z

given that:

$W_a \rightarrow$ for a solver with z_{AI}

$S_a \rightarrow$ for workers with z_{AI}

There is only uncertainty in:

$$(z) W_p \rightarrow S_p \quad (s = m(z))$$

Equilibrium:

For y workers

$$\int_y h(1-u) dG(u) = \int_{m(y)} dG(u)$$

Demand for Solver's time from the y workers, $m(y) \equiv$ set of solvers matched, Each has 1 unit of time

Demand = Supply

With this we characterize equilibrium as:

$$m_i + m_w + m_s = m \quad \text{All compute is assigned}$$

$$I \cup W_p \cup W_a \cup S_p \cup S_a = [0,1] \quad \text{All workers are occupied}$$

Each firm also chooses how to work